

STANNARD ROCK, MI, USA

WMO: 994200

Lat: 47.184N

Lon: 87.225W

Elev: 600

StdP: 14.38

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.3	4.1	-8.3	3.6	1.4	-4.3	4.5	5.5	62.8	26.1	59.0	26.2	26.0	290	1.023

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.9	75.6	67.2	73.0	66.1	70.8	64.8	70.5	72.8	68.2	71.4	66.3	69.8	20.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.7	111.7	71.6	67.1	101.7	69.6	65.0	94.6	68.1	34.8	73.0	32.8	71.3	31.3	69.8	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
48.8	42.0	37.2	-3.8	81.1	6.5	3.9	-8.4	83.9	-12.2	86.2	-15.8	88.4	-20.5	91.2		
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.9	20.7	20.4	28.1	37.0	46.1	54.7	61.7	63.4	58.6	47.2	36.2	27.6
	DBStd	16.72	9.19	9.17	9.26	6.85	7.07	6.54	6.35	5.19	6.27	6.79	7.72	7.39
	HDD50	4258	909	830	682	394	164	23	2	0	8	136	416	694
	HDD65	8523	1374	1250	1145	840	586	316	141	92	205	551	864	1159
	CDD50	1308	0	0	1	3	44	165	363	415	265	50	2	0
	CDD65	99	0	0	0	0	1	7	37	42	12	0	0	0
	CDH74	150	0	0	0	0	3	23	56	48	20	0	0	0
	CDH80	8	0	0	0	0	0	1	3	2	2	0	0	0
(14) Wind	WSAvg	20.2	23.6	23.5	20.6	19.8	19.6	17.2	16.7	16.3	18.8	21.3	22.8	22.9
(15) Precipitation	PrecAvg	28.3	2.1	1.3	1.5	2.2	3.0	2.7	2.4	2.1	3.1	3.0	2.2	2.3
	PrecMax	39.3	5.5	2.6	3.5	4.5	6.9	6.4	6.6	6.1	8.5	8.2	5.5	4.4
	PrecMin	19.4	0.5	0.4	0.0	0.6	1.3	0.5	0.9	0.4	1.0	0.8	1.0	0.3
	PrecStd	5.5	1.3	0.6	0.9	1.1	1.4	1.3	1.2	1.5	1.5	1.7	1.0	1.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.1	40.3	56.8	59.8	72.7	77.5	78.5	78.6	77.1	68.0	54.7	43.9
		MCWB	38.9	37.4	54.5	50.8	62.6	64.6	68.8	67.9	67.5	61.2	50.9	40.2
	2%	DB	38.2	36.3	47.1	53.8	66.3	72.2	75.3	75.0	71.3	63.0	50.3	41.2
		MCWB	36.0	33.9	41.8	46.6	57.7	63.2	67.3	68.9	64.5	57.0	46.7	38.8
	5%	DB	36.1	34.1	42.4	49.1	61.2	68.6	72.8	72.3	68.1	59.2	48.0	39.6
		MCWB	34.3	32.0	38.8	43.0	53.1	59.9	66.0	66.7	63.2	55.0	44.7	37.7
	10%	DB	33.4	32.1	38.9	45.2	57.1	65.4	70.7	70.1	66.1	56.4	46.0	37.7
		MCWB	31.5	29.9	36.0	39.8	50.0	58.7	65.0	65.0	61.9	53.1	43.2	36.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.3	38.1	55.6	54.4	67.4	69.2	73.7	73.4	69.0	63.5	51.1	41.5
		MCDB	41.1	40.0	58.8	59.8	71.4	75.5	76.5	75.0	73.6	65.2	53.7	41.9
	2%	WB	36.8	34.7	44.1	48.1	60.3	65.8	70.4	71.0	66.4	58.7	47.5	38.9
		MCDB	38.0	36.1	46.8	52.8	64.7	70.0	72.6	72.5	69.7	61.8	49.1	39.7
	5%	WB	34.8	32.5	39.8	44.3	55.3	63.1	68.3	68.1	64.4	55.8	45.1	37.3
		MCDB	36.2	34.1	42.5	49.0	59.3	68.0	71.1	71.4	67.0	58.3	46.7	38.6
	10%	WB	32.3	30.3	36.6	41.1	50.8	60.2	66.5	66.0	62.7	53.5	43.3	35.4
		MCDB	34.0	32.1	39.4	45.1	55.8	64.5	69.8	69.4	65.2	56.3	45.2	36.9
(35) Mean Daily Temperature Range	5% DB	MDBR	8.1	8.0	8.2	8.3	11.8	14.1	12.0	7.9	7.3	6.3	6.8	6.9
		MCDBR	9.1	9.4	11.5	14.6	20.1	17.7	13.5	10.5	10.3	10.0	8.2	8.2
		MCWBR	8.7	8.4	8.6	9.1	12.7	11.4	9.1	6.7	6.9	7.4	7.5	8.7
	5% WB	MCDBR	8.5	9.2	11.5	15.0	19.2	17.2	12.0	9.7	9.7	9.7	7.6	8.3
		MCWBR	8.7	9.0	9.5	10.6	14.8	13.5	9.9	7.7	7.8	8.3	7.3	8.5
(40) Clear-Sky Solar Irradiance	taub		0.278	0.291	0.314	0.344	0.379	0.385	0.395	0.375	0.367	0.335	0.305	0.273
	taud		2.512	2.458	2.413	2.369	2.370	2.376	2.396	2.453	2.458	2.515	2.524	2.543
	Ebn at Noon		263	281	289	289	281	279	274	276	267	261	249	252
	Edh at Noon		20	25	31	36	37	37	36	33	30	24	20	17
(44) All-Sky Solar Radiation	RadAvg		292	556	1040	1462	1777	1940	1961	1688	1211	703	342	220
	RadStd		22	68	127	165	119	143	102	115	71	87	53	25

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page